

Machine Shop 101 Bhadrata mariyu Parikarala Vaadaka Margadarsini

Mundu ga mukhyamainadi: Bhadrata.

Kallu motha pade shoes thappakunda vesukovali.

Safety glasses (bhadrata kanti addalu) thappakunda vesukovali. Nuvvu rendu second lu velthunna lada rendu gantalu velthunna anedi naku avasaram ledu. Safety glasses thappaka vesuko.

Nuvvu emi cut cheyatledu anna, akkada emi panulu nadavatledu anna, lada nuvvu vaarakae akkada nilabadi unna naku avasaram ledu, safety glasses vesuko.

Endukante? Evaraina nee venukaki vachi, machine on chesthe, oka aluminum mukkalu gaalilo egiri vachi ninna thagilavachhu.

Sadhana Avagahana

Nenu mummallandarini shop loki thesukupoyi, prathi parikaranni chupinchi, samagri ni sariga ela clamp cheyalo vivaristhanu. Ee parikarallo deninaina vaadadaniki clamping anedi chala mukhyamaina bhagam. Nenu maku chupinchaka, meeru swayam ga prayatnistaru. Meeru sariga cheyagalithe, meeru tayarugannatle.

Maku podavu juttu unte, venakaki mudi veyandi. Inthaku mundu konni sangathanalu jarigayi, oka ammayi juttu drill press lo irukkupoyindi. Adi chala pramanakaramaina paristhiti, amaye juttu kathinchalsi vachestunde.

Shop lo unna Machine lu

- Belt sander
- Table saw
- Band saw
- Horizontal band saw
- Compound miter saw
- Drill press
- Power drills (wire unnavi mariyu wire lenivi)
- Lathe
- Mill
- CNC

- Water Jet
 - Laser Engraver
 - 3D Printer
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Band Saw

Idhi oka sadharana band saw. Meelo chalamandiki e rakam alavatu unde untundhi: rendu chakrala meedha thirige blade. Idhi off lo unnapatiki, eppudu mee vellu lopala pettakandi. Charcha loni loppala valana machine lu porapatuna switch on avvadam nenu choosanu.

Meeru dhintho ekkuvaga aluminum tubing, shafts, mariyu Lexan sheets cut chestharu. If you're cutting something small, the vertical band saw is your go-to, not the horizontal or compound miter saw.

Cut chesthunnapudu:

- Thinnani cut la kosam guide ni vaadandi
- Samagri ni kolichi, gattiga pattukondi
- Mee vellani blade ki dhooranga unchandi

Meeru samagri ni blade loki thosthunnaru kabatti, dheenipai clamp ni vaadaru.

Sadharana niyamam: Idhi 1/8" thikkiness kante ekkuva unte, mellaniga vellandi. Meeru veganga velthe, blade moosukupothundhi, ninnhadhisthundhi, mariyu meeku aa vithinthina kalina vaasana vasthundhi. Dhanni nivarinchandi.

Power Drill

Cordless ayina corded ayina, rendu upayogakarangane untayi. Corded avi chala strong ga untayi kani vatiki chuck key avasaram. Cordless vatiki torque settings maruchukone saulabhyam untundi.

Screws leda bolts ni "strip" cheyakunda undataniki takkuva torque vaadandi. Screw "strip" avvadam ante meeru dani head ni entha ksaayincharu ante inka e bit kooda saripodu.

Daaniki badulugaa,

- Dani bayataku drill cheyadam
- Vice grips vaadadam
- Flathead ga marustaru (paryapthamaina samagri migili unte)

Prathi sari evaraina screw strip cheste, meeru naku 50 cents ivvali. :)

Bit ni lopala peduthunnappudu, ivvi choosukondi:

- Idi poorthiga kindaku tostharu
- Idi sariga kashtharu

Lekapothu, drill chesthunnappudu idi bayataku egiri povachhu.

Horizontal Band Saw

Idi chala bhadramaina machine laloni okati.

Idhi horizontal ga leda vertical ga vaadavachhu. Kani mentor OK ani cheppenthavaraku dhanni vertical ga lock cheyakandi.

Dheeniki built-in clamp untundi, samagri ni unchi, clamp chesi, taravatha blade ni mellaga kindaku dinchandi.

Ee parikaram deniki manchidi:

- Mande manapina samagri ni cut cheyadaniki
- Podavaina, saafugaanunna, thinnani cuts kosam

Idi ninnhadhiga anipisthe, meeru oka flat surface ni thaguluthunnaru mariyu okkasariga ekkuva teeth pani chesthunnayi ani artham. Orupuga undandi, idi poorthiga cut chesi daanitho adhe off avuthundi.

Compound Miter Saw

Shop lo unna parikaralalo idhe chala pramanakaramainadi.

Meeru "ow" ani ane munde ee machine mee velu ni theesesthundi. Inthaku mundu oka vyakthi vellalo rendu steel rods tho muginchalsi vachindi. Parikaranni gauravinchandi.

Deeni kosam dheenini upayoginchandi:

- Thinnani, chaturatram cuts
- Mandamaina samagri
- Podavaina aluminum bars vanti stock material

Prathidheenini clamp cheyandi. Chinna mukkalu viseshanga pramanakaramainavi. Avi bayataku egiri povachhu.

F-style clamps leda mee daggara unnavi vaadandi, muka kadalaleani nischayam chesukondi.

Cutting linku sambandhinchu meru ekkada cut chestunnaro gamaninchandi. Endukante blade kontha padardhaanni tolagistundi (deenini “karf”), deenivalla mee bhagam antha kaccitanga cut avvakapovachu.

Dheenni vaaduthunnappudu:

1. Trigger ni lagandi
2. Mellaga material ni cut cheyandi
3. Meeru mukkani clear chesaka matrame vadhilandi, lopala unna samayamlo kadhu

Pani poorthayyakatha eppudoo blade ni kindaku lock cheyandi. Mariyu shop nunchi velleppudu unplug cheyandi.

Jigsaw

Dheenni cut cheyadaniki vaadandi:

- Aluminum sheets
- Lexan
- Plywood
- Drywall

Idi drills vaade ade batteries tho nadusthundhi. Material ni batti meeru cutting mode ni adjust chesukovachhu. Aluminum kosam Mode 0, Lexan kosam Mode 1 vaadandi.

Dheenni oka baha meeda vaadandi. Kindaku tostharu, paikikadhu. Idhi vibration ni thaggisthundhi mariyu shubhramaina cut ni isthundhi.

Pedda sheets kosam C-clamps mariyu chinna mukkala kosam F-style clamps vaadandi.

Drill Press

Drill press ni dheeni kosam vaadandi:

- Thinnani, kachithamaina thutalu/holes
- Bearings mariyu kachithamaina bhagalu

Samagri ni mee chethulatho pattukovadhu, drill vice vaadandi. Meeru chinna bhagalani drill chestunnatlaithu, vatinni clamp cheyandi.

Drill bits:

- Pedda bits tho mellaniga vellandi
- Chinnaga parimanamtho praarambinchi, kramamga penchukuntu vellandi. (Ex. 9/16" angulalato praarambinchavaddu.)

Bit entha lothuku vellalo daniki taggattu depth stop ni adjust cheyandi. Prarambhinchadaniki munde bit lock ayindhi ani mariyu height clamp bhadramga undhi ani nischayam chesukondi.

De-Burring

Drill chesina thutalalo vaadaina anchulu (burrs) migilipovachhu. Inspector mee robot meeda tana velu cut chesukogaligithe, meeru inspection pass kavu. Dhanni chethitho shubhram cheyadaniki deburring tool leda pedda drill bit vaadandi.

Sanders

Manam sanders ni dheeni kosam vaadatham:

- Deburring
- Chinna motthamlo material ni theeseyadaniki

Pedda motthamlo theeseyadaniki sanders vaadakandi. Idi belts ni moosivesthundhi leda thenchusthundhi, viseshanga aluminum tho.

Finishing kosam sanders vaadandi:

- Anchaukaḷai konchem subhram cheyadam.
- Moolalanu round cheyadam
- Cuts ni saafu cheyadam

Eppudoo evarikaina screwdriver leda vaadaina vasthuvunu thula vaapu ivvakandi. Eppudoo handle vaapu ivvandi.

Dremel

Dremel stationenu veeti kosam upayogistaru:

- Sannani sanding

- Mini-drilling
- Anchulu shubhram cheyadam

Chetito pattukune Dremels kuda andubatulo unnayi

Bhadrata meeda Akharu Suchanal

- Safety glasses thappakunda vesukovali
- Podavu juttu unte venakaki mudi veyali
- Edaina vishayampai meeku evaina sandehalu ledaa anischiti unte. Tappakundaa mee upadhyayudini adagandi.
- Meeru sariga ela vaadalo thelithe thappa eppudoo e parikaranni vaadakandi

Meeru parikaralanu choosi mariyu prayathninchina tharavatha, meeku theliyali:

- Idi deni kosam
- Mee samagri ni ela clamp cheyali
- Dhanni bhadramga ela vaadali

Idi paripurnamga undaalani aashinchadam ledi. Ayite, meru surakshitamgaa untarani mariyu edaina teliyakapote prasnalu aduguthaarani nenu aasistunna.

Manamandarim niyamalanu paalisthe, manam bhadramga, samardhavangaga nirminchavachhu, inspection pass avvavachhu, mariyu tondaraga field loki vellavachhu.